

Plant Health Detection System

AI-Powered Analysis Platform

RGB Imaging Solution

Technical Proposal

Starting with Sugar Maple → Expanding to Multiple Species

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Sensor: RGB Camera

1 Overview

This proposal outlines an AI-powered plant health detection system using RGB drone imagery. The system will be developed in phases, **starting with Sugar Maple as the base model**, then expanding to support multiple plant species.

Phase 1: Sugar Maple → Phase 2+: Additional Species

1.1 Problem Statement

Traditional plant health assessment relies on manual inspection, which is:

- **Time-intensive:** Ground-based surveys cover limited area per day
- **Subjective:** Assessments vary between inspectors
- **Reactive:** Problems often detected after significant damage occurs
- **Costly:** Requires trained arborists for accurate diagnosis

1.2 Proposed Solution

An automated AI system that analyzes drone imagery to provide:

- **Scalable coverage:** Analyze hundreds of trees per flight
- **Consistent results:** Standardized detection criteria across all assessments
- **Early detection:** Identify stress indicators before visible decline
- **Actionable reports:** Prioritized recommendations for intervention

2 Detection Modules

The system comprises five detection modules that work together to provide comprehensive plant health assessment.

2.1 Module 1: Species Detection

Integration with 3rd party plant identification API to detect and confirm species before analysis.

Upload Image → API Call → Species Confirmed → Route to Species Model

Integration Details:

- **API Provider:** PlantNet or similar botanical identification service
- **Confidence threshold:** $\geq 85\%$ required for automatic routing
- **Fallback handling:** Manual review queue for low-confidence identifications
- **Caching:** Species confirmed once per tree location to reduce API calls

Supported Species (Phase 1):

Common Name	Scientific Name
Sugar Maple	<i>Acer saccharum</i>

2.2 Module 2: Maturity Classification

Classify plants into growth stages based on crown/canopy characteristics.

Stage	Detection Method
Seedling	Small crown, simple structure
Sapling	Medium crown, developing canopy
Mature	Large crown, full canopy density

Technical Approach:

- Mask R-CNN for individual crown segmentation
- Photogrammetric height estimation from drone imagery
- Crown diameter and density analysis

Classification Thresholds (Sugar Maple):

Stage	Height	Crown Diameter	DBH Est.
Seedling	<2m	<1.5m	<5cm
Sapling	2–8m	1.5–5m	5–15cm
Mature	>8m	>5m	>15cm

Expected Accuracy: 90%+ classification accuracy based on crown characteristics visible in RGB imagery at 2–3cm GSD (Ground Sample Distance).

2.3 Module 3: Leaf Disease Detection

Detect and classify common foliar diseases using pattern recognition.

For Sugar Maple (Phase 1):

Disease	Visual Indicator	AI Method
Tar Spot	Black spots on leaves	Object detection (YOLO)
Anthrachnose	Brown necrotic patches	Texture + color analysis
Leaf Spot	Circular brown lesions	Pattern matching
Powdery Mildew	White powdery coating	Albedo change detection

Expansion to Other Species:

- Train additional disease classes per species
- Modular architecture allows adding new disease types
- Transfer learning from base model speeds up training

Severity Scoring:

Level	Score	Criteria
None	0	No visible disease symptoms
Light	1–2	<10% of visible foliage affected
Moderate	3–4	10–30% of visible foliage affected
Severe	5	>30% of visible foliage affected

Output Format:

- Disease type with confidence score (0–100%)
- Affected area percentage estimate
- Severity rating (0–5 scale)
- Bounding boxes on detected lesions/symptoms

2.4 Module 4: Vascular Disease Indicators

Detect visual symptoms suggesting internal vascular problems.

For Sugar Maple (Phase 1):

Condition	Visible Symptoms Detected
Verticillium Wilt (Suspected)	Branch wilting, crown dieback, leaf browning
Eutypella Canker (Suspected)	Trunk deformation, bark anomalies

Detection Approach:

- Crown sector analysis for asymmetric dieback
- Temporal comparison across survey dates
- Oblique imagery for trunk-level symptoms

Important Limitations:

- RGB imagery detects *symptoms*, not pathogens directly
- Results labeled as “Suspected” require ground verification
- Internal vascular damage may not show external symptoms until advanced stages

Recommendation Output:

Confidence	Recommended Action
High (>80%)	Priority ground inspection within 2 weeks
Medium (50–80%)	Schedule ground inspection within 1 month
Low (<50%)	Monitor in next survey cycle

2.5 Module 5: Damage Detection

Identify structural damage visible in RGB imagery.

Damage Type	Detection Method
Dead Limb	Color difference (brown/gray vs green)
Broken Limb	Geometric anomaly in crown structure
Trunk Cavity	Visible external damage in oblique shots
Crown Asymmetry	Shape analysis indicating structural loss
Storm Damage	Multiple indicators combined detection

Risk Assessment Output:

Risk Level	Description
Low	Minor cosmetic damage, no structural concern
Medium	Moderate damage requiring monitoring
High	Significant structural damage, recommend arborist inspection
Critical	Imminent hazard, immediate attention required

2.6 Module 6: Size Estimation (Optional)

Estimate physical dimensions using photogrammetry.

Metric	Method
Height	Digital Surface Model from drone imagery
Crown Width	Segmentation boundary measurement
Trunk Diameter	Allometric estimation from height + crown
Crown Volume	3D reconstruction from overlapping imagery

Expected Accuracy:

Metric	Expected Accuracy	Requirements
Height	$\pm 0.5\text{m}$	RTK-enabled drone, GCPs
Crown Width	$\pm 0.3\text{m}$	2cm GSD minimum
Trunk Diameter	$\pm 15\%$	Allometric model calibration

Note: Trunk diameter (DBH) estimation uses species-specific allometric equations. Accuracy improves with local calibration data.

3 Data Requirements**3.1 Training Data**

For optimal model performance, the following data is required:

Category	Minimum Images	Recommended
Healthy trees (various stages)	200	500+
Each disease type	100	300+
Damage examples	50	150+
Seasonal variation	2 seasons	4 seasons

3.2 Image Specifications

Parameter	Requirement
Ground Sample Distance (GSD)	1–3 cm/pixel recommended
Flight altitude	30–60m AGL (species dependent)
Overlap	75% frontal, 65% side minimum
Camera	RGB, minimum 20MP
Format	JPEG or TIFF, geotagged
Lighting	Overcast preferred, avoid harsh shadows

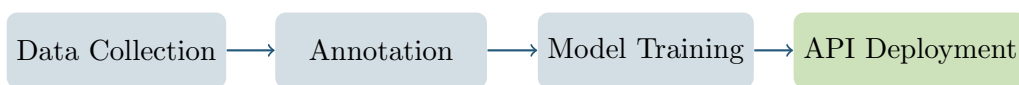
3.3 Annotation Requirements

- **Bounding boxes:** Individual tree crown delineation
- **Polygons:** Disease lesion boundaries for training
- **Labels:** Species, maturity stage, disease type, severity
- **Metadata:** Date, location, ground truth verification where available

Client Responsibility: Provide raw drone imagery and any existing ground truth data.

4 Process Flow

4.1 Phase 1: Sugar Maple Base Model

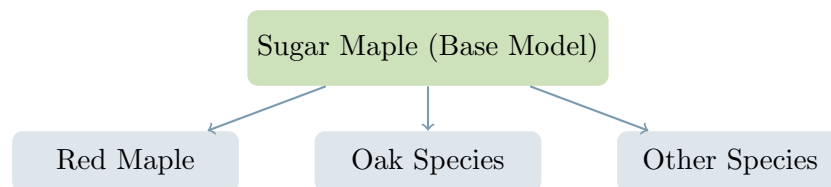


Phase 1 Milestones:

Milestone	Week	Deliverable
M1: Data Ready	W3	Annotated dataset complete
M2: Models Trained	W6	All modules achieving target accuracy
M3: API Deployed	W8	Functional API in staging environment
M4: Final Delivery	W10-12	Production-ready system with documentation

4.2 Phase 2+: Multi-Species Expansion

Once the Sugar Maple base model is complete, the system can expand to additional species.



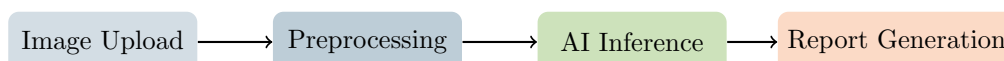
Transfer learning reduces training time for new species

Expansion Process for Each New Species:

1. Collect species-specific training images
2. Annotate species-unique diseases and conditions
3. Fine-tune base model with new data
4. Validate and deploy updated API

5 Technical Architecture

5.1 System Components



5.2 Technology Stack

Component	Technology
Deep Learning Framework	PyTorch / TensorFlow
Object Detection	YOLO, Mask R-CNN
Image Processing	OpenCV, Pillow
API Framework	FastAPI (Python)
Database	PostgreSQL with PostGIS
Cloud Infrastructure	AWS / Azure / GCP (client choice)
Containerization	Docker

5.3 API Specification

Endpoints:

- POST /analyze – Submit image for full analysis
- GET /results/{id} – Retrieve analysis results
- POST /batch – Submit multiple images for batch processing
- GET /health – Service health check

Response Format: JSON with structured results per module, including confidence scores, bounding boxes, and severity ratings.

Authentication: API key-based authentication with rate limiting.

6 Team Structure

Role	Count	Responsibilities
ML Engineer (Lead)	1	Architecture design, model optimization, API development
ML Engineer	1	Model training, evaluation, deployment support
Annotation Lead	1	Quality control, annotation guidelines, team coordination
Data Annotators	2	Image labeling, bounding box creation, data validation
Total	5	

7 Quality Assurance

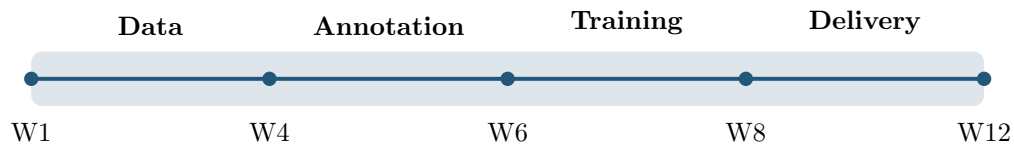
7.1 Testing Protocol

1. **Unit Testing:** Individual model component validation
2. **Integration Testing:** End-to-end pipeline verification
3. **Field Validation:** Testing on new imagery not used in training
4. **Expert Review:** Arborist validation of sample outputs

7.2 Continuous Improvement

- Feedback loop for misclassifications to improve model
- Quarterly model retraining with accumulated data
- Version control for all model iterations

8 Timeline



9 Investment

Module	Cost (USD)
Species Detection (3rd Party API)	Included
Maturity Classification	\$3,500
Leaf Disease Detection (4 types)	\$4,500
Vascular Disease Indicators	\$3,500
Damage Detection	\$2,500
Size Estimation	\$1,000
Total (Phase 1 - Sugar Maple)	\$15,000

Timeline: 8-12 Weeks

Payment Terms: 30% Advance
— 40% Midpoint — 30% Delivery

Future Species Expansion: Pricing for additional species will be quoted separately based on disease complexity and annotation requirements. Transfer learning from the base model significantly reduces cost for subsequent species.

10 Deliverables

Upon project completion, the following will be delivered:

1. Trained AI Models

- Maturity classification model (Sugar Maple)
- Leaf disease detection model (4 disease types)

- Vascular disease indicator model
- Damage detection model
- Size estimation model

2. REST API Service

- Dockerized deployment package
- API documentation (OpenAPI/Swagger)
- Sample client code (Python, JavaScript)

3. Documentation

- System architecture documentation
- Model training methodology
- Integration guide
- User manual

4. Support

- 30-day post-deployment support
- Bug fixes during support period
- Knowledge transfer session (2 hours)

11 Assumptions & Dependencies

11.1 Assumptions

- Client will provide sufficient training imagery meeting specified requirements
- Drone imagery will be captured following recommended flight parameters
- Access to subject matter expert (arborist) for validation queries
- Stable internet connection for API deployment and testing

11.2 Dependencies

- Third-party species identification API availability and pricing
- Cloud infrastructure provisioning (client or provided)
- Timely feedback on validation results during development

11.3 Out of Scope

- Drone hardware or flight operations
- Multispectral or hyperspectral imagery analysis
- Real-time streaming video analysis
- Mobile application development
- Root disease detection (not visible in aerial imagery)

12 Next Steps

1. **Proposal Acceptance:** Review and sign-off on scope and pricing
2. **Data Assessment:** Evaluate existing imagery and identify gaps
3. **Kickoff Meeting:** Align on timeline, communication, and milestones

For questions or clarifications, please contact us.